# Statistical Modeling of Housing Prices Using Spatial and Structural Features

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## 1 Introduction

Housing markets are influenced by a combination of structural property characteristics, accessibility to transportation, neighborhood amenities, and geographic location. Understanding how these factors interact is important for both economic analysis and urban planning, as housing prices often reflect broader patterns in infrastructure development.

The goal of this analysis is to understand how structural, temporal, and spatial characteristics of properties affect housing prices. The dataset contains information about historical real estate valuations in the Sindian District of New Taipei City, Taiwan, including variables describing house age, proximity to transportation infrastructure, neighborhood amenities, and geographic location. By applying statistical modeling techniques, this analysis aims to identify which features are most strongly associated with housing price variation and evaluate how modeling choices influence predictive performance.

## 2 Problem Statement

Housing price prediction is an important problem in real estate economics and urban planning. Accurate modeling of housing prices can help identify key drivers of property value, guide investment decisions, and provide insight into how accessibility, neighborhood characteristics, and spatial location influence real estate markets.

However, relationships between housing prices and property features are not always strictly linear. Some variables may have nonlinear effects, and spatial variables may introduce correlation that complicate standard regression modeling. As a result, careful statistical modeling and analysis are necessary to ensure that the fitted models provide reliable insights into the factors affecting housing prices.

The primary goals of this project are:

1. Identify which predictors are significantly associated with housing price per unit area.
2. Evaluate whether nonlinear relationships exist between predictors and the response.
3. Improve statistical models through feature transformations and dimensionality reduction.
4. Evaluate uncertainty in coefficient estimates using bootstrap methods.
5. Examine classification performance by modeling whether housing prices exceed a threshold using logistic regression.

Preprint. Under review.

## 3 Data Description

The dataset used in this analysis is the Real Estate Valuation dataset[1], which contains information on housing transactions in Sindian District, New Taipei City, Taiwan. The dataset includes several explanatory variables related to property characteristics, location, and accessibility. The response variable is house price per unit area, measured in units of 10,000 New Taiwan Dollars per Ping, roughly equivalent to $8.83 per square foot.

The dataset contains 414 observations with one response variable and six predictor variables. Table 1, seen below, shows the variables along with a short description of each one.

| **Variable** | **Description** | **Type** |
|---|---|---|
| X1 | Transaction date | Nonnegative Continuous |
| X2 | House age (years) | Nonnegative Integer |
| X3 | Distance to nearest MRT station (meters) | Nonnegative Continuous |
| X4 | Number of convenience stores nearby | Nonnegative Integer |
| X5 | Latitude | Continuous |
| X6 | Longitude | Continuous |
| Y | House price per unit area | Nonnegative Continuous |

Table 1: Variable Information

The predictors in the dataset represent important determinants of housing prices, including property age, accessibility to transportation, neighborhood amenities, and geographic location. The geographic variables, latitude and longitude, indicate that all observations come from a relatively small area within New Taipei City, suggesting that spatial information may influence property values.

## 4 Methodology

For our analysis, we followed a specific methodology to evaluate the relationship between housing prices and structural features as well as spatial features.

1. We first conducted exploratory data analysis to understand the variables and identify potential nonlinear relationships between predictors and the response variable. This included graphical analysis of histograms, bar plots, and pairwise scatter plots.
2. Next, we fit a baseline multiple linear regression model using all the available predictors. To evaluate performance, diagnostic plots were used to assess any violations of model assumptions including nonlinearity and heteroscedasticity.
3. After viewing the baseline results, we used feature engineering techniques to address the observed violations. Specifically, we used nonlinear transformations and dimensionality reduction through principal components analysis. We then fit a new refined regression model using these transformations and used residual diagnostics and model summary statistics to evaluate.
4. Finally, bootstrap sampling was used to estimate the variability of coefficient estimates, and a logistic regression model was fitted to evaluate classification performance when predicting whether housing prices exceed the median value.

## 5 Exploratory Data Analysis

Before fitting statistical models, we conducted exploratory data analysis to better understand the distributions of the variables and the relationships between predictors and the response variable. By visualizing the data, we can identify skewness, potential outliers, and nonlinear relationships that may influence model selection.

[1]The Real Estate Valuation dataset comes from the UC Irvine Machine Learning Repository.

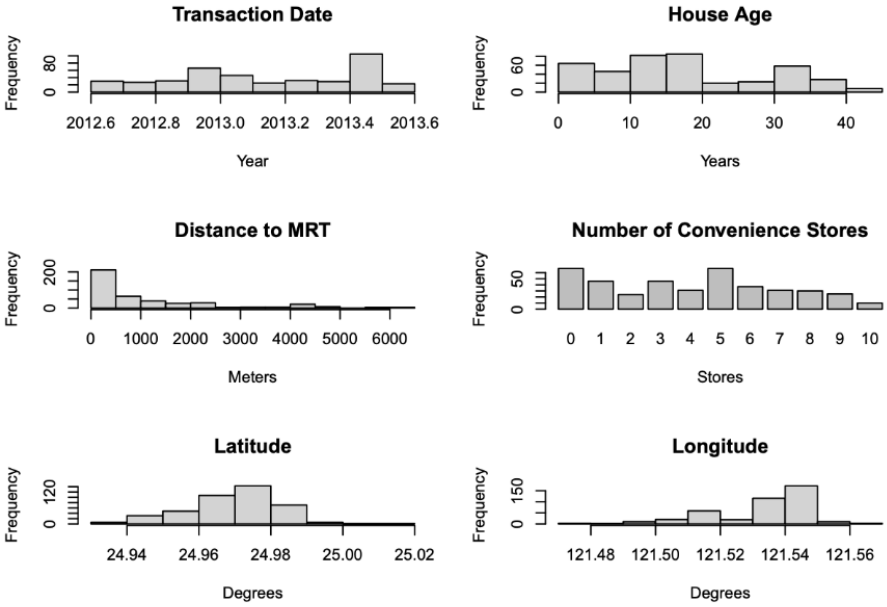



Figure 1: Distribution of Predictor Variables.

Figure 1 shows the marginal distributions of the predictor variables. Transaction dates appear relatively evenly distributed across the observation period, with a noticeable concentration of transactions occurring around early 2013. House age displays a right-skewed distribution, indicating that a large portion of the properties in the dataset are relatively new while fewer observations are older houses.

Distance to the nearest MRT station is strongly right-skewed, suggesting that many properties are close to transit stations while many fewer are located farther away. The number of nearby convenience stores shows a roughly balanced distribution across categories. The geographic coordinates have a very narrow range, which confirms that the properties are located within a small region.

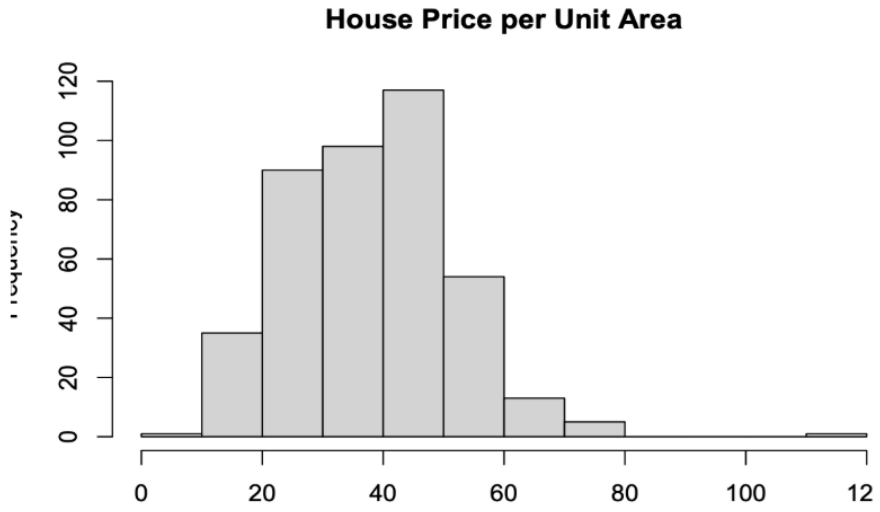



Figure 2: Distribution of Response Variable.

Figure 2 shows the distribution of the response variable, house price per unit area. The response exhibits moderate right-skewness with a small number of high-price observations.

To examine relationships between variables, pairwise scatter plots were also constructed[2]. The plots suggest several potentially nonlinear relationships, particularly between house age and housing price as well as distance to the nearest MRT station and housing price. These patterns motivate the feature transformations introduced later in this report.

# 6 Analysis

We first fit a multiple linear regression model with house price as the response variable and all predictors included as covariates:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + \epsilon. \tag{1}$$

## 6.1 Assumptions of the Linear Regression Model

Classical linear regression relies on several assumptions to ensure valid statistical inference. These include linearity between predictors and the response, independence of the error terms, constant variance of the errors (homoscedasticity), approximate normality of the error distribution, and the absence of severe multicollinearity among predictors.

We checked these assumptions using residual diagnostics from the baseline model, which are shown in Figures 3 and 4. The model summary output revealed potential violations of the linearity assumptions[3]. This was further supported through closer inspection of the residual plots. In particular, residual plots showed curvature with respect to several predictors, indicating that the relationship between predictors and the response may not be strictly linear as well as potential heteroscedasticity in the model residuals. Figure 3 displays the model residuals as well as the residuals against every predictor variable. Additionally, Figure 4 contains the partial residual plots of all the predictors.

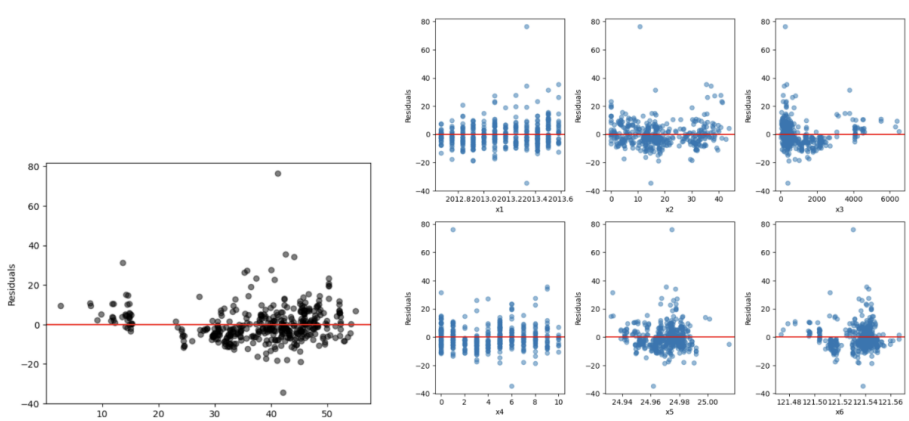



Figure 3: Original Model Residuals (Left) and Predictor Residuals (Right).

[2]See Appendix A for the pairwise scatter plots.
[3]See Appendix B for the model summary.

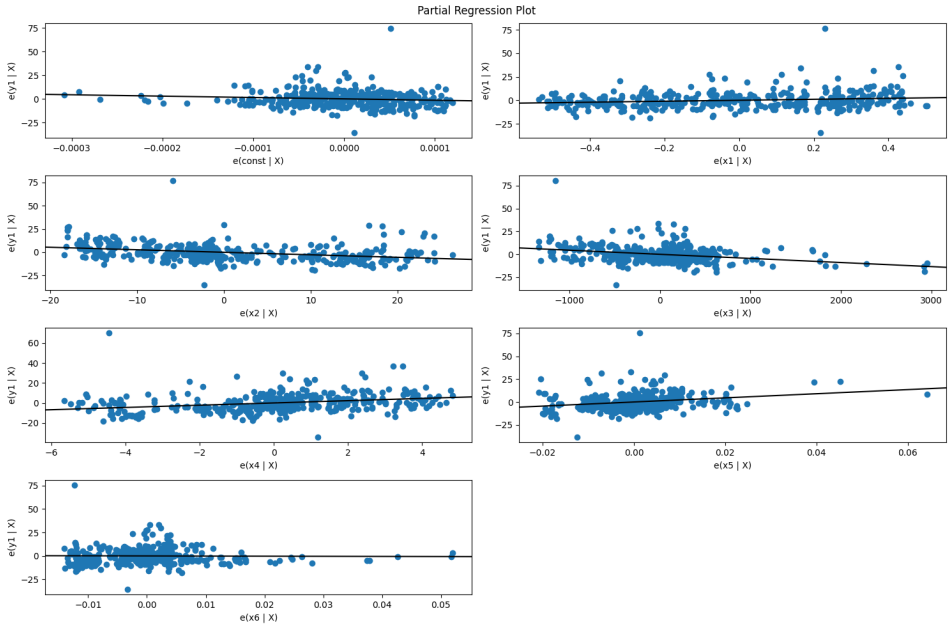



Figure 4: Original Model Partial Residual Plots.

### 6.2 Feature Engineering

These results suggest that the model could benefit from nonlinear transformations of certain predictors. In fact, to address the nonlinearities observed in the initial model, we introduced several feature transformations. Below we have included table 2, which summarizes the transformations we performed.

| Variable | Transformation |
|---|---|
| X2 | $(x_2)^2$ |
| X3 | $\log(x_3 + 1)$ |
| X5, X6 | Principal Component Analysis |

Table 2: Variable Transformations

The transformations above were motivated by patterns observed in the exploratory analysis.

First, we used a quadratic term for the house age. Housing prices decline as properties age due to depreciation and maintenance requirements, but the relationship is rarely strictly linear. Including a squared term allows the model to capture the curvature that we can see in the relationship between property age and housing value.

Second, the distance to the nearest MRT station was log-transformed using $\log(x_3 + 1)$. The original value displayed a strong right-skewed plot and large variation between nearby and distant properties. Log transformations compress the scale of large values and produce relationships that are closer to linear.

Finally, we used principal component analysis (PCA) was applied to the geographic variables (latitude and longitude). Because these coordinates are highly correlated and both capture spatial location, PCA summarizes the geographic variation into orthogonal components that capture the major directions of spatial variation. This reduces potential multicollinearity while retaining the spatial information from the original variables.

The refined model substantially improved the fit and produced more interpretable relationships between predictors and the response. It also helps address violations of the linearity assumption observed in the baseline model. The adjusted $R^2$ of the refined model increased by about 0.08 in

comparison to the baseline model, seen in the model summary output[4]. Figure 5, shown below, displays the residuals of the new model against all the predictors and Figure 6 contains the updated partial residual plots. The new residual plots now suggest homoscedasticity.

### 6.3 Refined Model Specification

Based on the diagnostic results and exploratory analysis, we constructed a refined regression model incorporating nonlinear transformations and spatial dimensionality reduction. The refined model is written as:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_2^2 + \beta_4 \log(x_3 + 1) + \beta_5 x_4 + \beta_6 PC_1 + \epsilon \tag{2}$$

where $PC_1$ represents the first principal component derived from the geographic variables latitude and longitude.

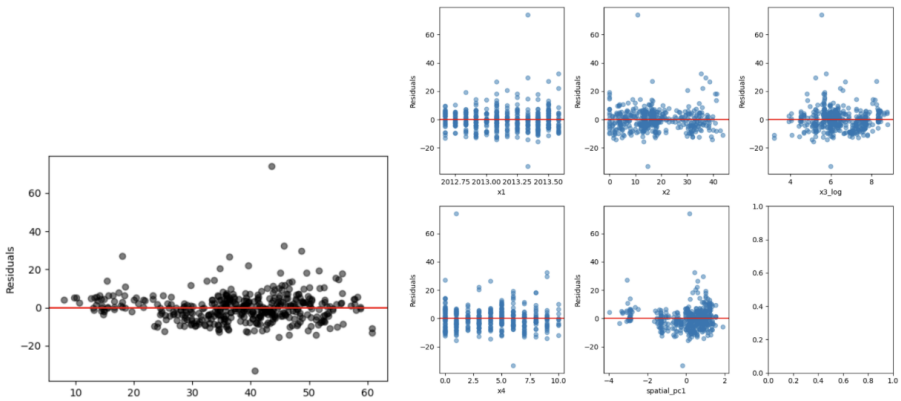



Figure 5: Refined Model Residuals (Left) and Predictor Residuals (Right).

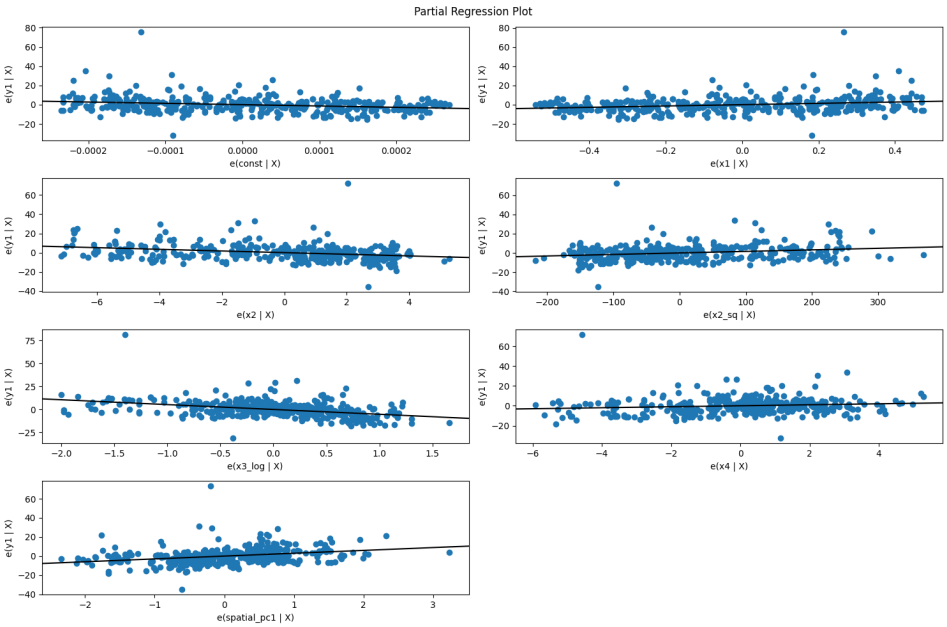



Figure 6: Refined Model Partial Residual Plots.

[4]See Appendix C for the refined model summary.

### 6.4 Model Fit Evaluation

The residual diagnostics suggest that the refined model provides a better fit to the data compared to the baseline model. In particular, the residual versus fitted plot shows a more random scatter around zero, indicating that the nonlinear patterns present in the original model have been mostly addressed.

Additionally, the spread of residuals appears more consistent across the range of fitted values, suggesting improved homoscedasticity. While some mild deviations remain, the refined model substantially improves the overall model assumptions and provides a more reliable representation of the relationship between predictors and housing prices.

### 6.5 Bootstrap Inference

To evaluate the uncertainty of the estimated regression coefficients, bootstrap resampling techniques were applied. The bootstrap is a nonparametric method that approximates the sampling distribution of an estimator by repeatedly resampling from the observed dataset.

Two bootstrap approaches were implemented: the case bootstrap and the residual bootstrap. In the case bootstrap, observations are resampled with replacement from the original dataset, and the regression model is refitted for each bootstrap sample. This method captures variability arising from both the predictors and the response variable.

On the other hand, the residual bootstrap keeps the predictor values fixed and resamples the model residuals. New response values are generated by adding resampled residuals to the fitted values from the original model. The regression model is then refitted to the simulated data.

For both bootstrap procedures, 5000 bootstrap samples were generated and the regression coefficients were recomputed for each sample. The variability of these estimates provides an empirical approximation of the standard errors of the coefficients, allowing us to assess the stability of the model estimates.

| Variable | Estimate | Std. Error |
|---|---|---|
| Intercept | -13,790 | 2858.188 |
| x1 | 6.8888 | 1.420 |
| x2 | -0.8600 | 0.139 |
| x3 | 0.0159 | 0.003 |
| x4 | -5.2678 | 0.591 |
| x5 | 0.4984 | 0.189 |
| x6 | 2.9720 | 0.448 |

| Variable | OLS SE | Residual Bootstrap SE | Case Bootstrap SE |
|---|---|---|---|
| Intercept | 2858.188 | 2878.590 | 2708.362 |
| x1 | 1.420 | 1.430 | 1.346 |
| x2 | 0.139 | 0.139 | 0.149 |
| x2_sq | 0.003 | 0.00335 | 0.00378 |
| x3_log | 0.591 | 0.593 | 0.772 |
| x4 | 0.189 | 0.188 | 0.252 |
| spatial_pc1 | 0.448 | 0.442 | 0.407 |

Figure 7: Coefficients and Associated Standard Errors of Original Model (Left) and Refined Model (Right).

The bootstrap results suggest that the coefficient estimates are relatively stable across resampled datasets, suggesting that the refined model provides robust estimates of the relationships between predictors and housing prices.

### 6.6 Model Comparison

Comparing the baseline and refined models highlights the importance of feature transformations in improving model performance. The refined model achieved a substantially higher adjusted $R^2$ compared to the baseline linear model, indicating that the transformed predictors explain a larger proportion of the variation in housing prices.

In addition to improved goodness-of-fit, the refined model exhibits better diagnostic behavior in the residual plots. These improvements suggest that incorporating nonlinear transformations and spatial dimensionality reduction results in a more appropriate statistical model for the data.

The bootstrap standard errors were very similar to the classical OLS standard errors. Residual bootstrap estimates were nearly identical to the classical estimates, while case bootstrap standard errors were slightly larger for some predictors. These results suggest that the classical OLS inference is relatively robust for this dataset. Furthermore, to explore classification performance, we constructed

a logistic regression model by converting the continuous response variable into a binary outcome. The response was binarized using the median housing price as a threshold. Using the median ensures a roughly balanced classification problem. Properties with prices above the median were labeled as 1, and those below the median were labeled as 0. The logistic regression model uses the same predictors as the refined linear model and is written as:

$$\log\left(\frac{P(Y=1)}{1-P(Y=1)}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 \log(x_3+1) + \beta_4 x_4 + \beta_5 PC_1. \tag{3}$$

Here, $P(Y=1)$ represents the probability that a property belongs to the high-price category (above the median housing price).

Figure 8, shown below, displays the model summary output.

```
                           Logit Regression Results
==============================================================================
Dep. Variable:                  y_bin   No. Observations:                  414
Model:                          Logit   Df Residuals:                      407
Method:                           MLE   Df Model:                            6
Date:                Thu, 26 Feb 2026   Pseudo R-squ.:                  0.4925
Time:                        04:24:18   Log-Likelihood:                -145.64
converged:                       True   LL-Null:                       -286.96
Covariance Type:            nonrobust   LLR p-value:                 4.284e-58
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const       -2657.0938   1114.779     -2.384      0.017   -4842.021    -472.167
x1              1.3251      0.554      2.392      0.017       0.239       2.411
x2             -0.1571      0.056     -2.804      0.005      -0.267      -0.047
x2_sq           0.0023      0.001      1.873      0.061      -0.000       0.005
log_x3         -1.5159      0.247     -6.140      0.000      -2.000      -1.032
x4              0.1208      0.065      1.846      0.065      -0.007       0.249
spatial_pc1     1.2260      0.261      4.690      0.000       0.714       1.738
==============================================================================
```

Figure 8: Logistic Regression Model Summary.

### 6.7 Interpretation of Model Coefficients

The refined regression model provides insight into how different features influence housing prices. The coefficient for the number of nearby convenience stores is positive, indicating that each additional convenience store nearby is associated with an increase of approximately $\beta_4$ units in housing price per unit area, holding other variables constant. This finding is consistent with expectations, as access to services and retail is an attractive feature for residents. The distance to the nearest MRT station has a negative relationship with housing prices, suggesting that properties located farther from public transportation tend to have lower values. This reflects the importance of accessibility in urban housing markets. The spatial principal component captures geographic variation in housing prices, indicating that location within the district plays a significant role in determining property values. This spatial effect highlights the importance of neighborhood-level differences even within relatively small geographic regions.

The results suggest that transaction date and spatial location strongly influence whether a property falls above the median price, house age and distance to MRT tend to decrease the probability of belonging to the high-price category, and neighborhood amenities such as convenience stores are positively associated with higher prices. Overall, the model highlights the importance of accessibility and spatial variation in housing valuation. To identify a reduced model, we applied LASSO logistic regression with cross-validation. This approach performs variable selection by shrinking less important coefficients toward zero. After performing LASSO logistic regression, the reduced model retained x1, x2, log(x3+1), x4, and the spatial principal component. The term $(x2)^2$ was not selected. After selecting predictors, bootstrap resampling was used again to estimate the standard errors of the coefficients in the reduced model. The results are shown in Figure 9, below.

| Variable | Bootstrap Std. Error |
|---|---|
| Intercept | 1534.834 |
| x1 | 0.504 |
| x2 | 0.064 |
| log(x3+1) | 0.304 |
| x4 | 0.065 |
| spatial_pc | 20.351 |

Figure 9: Coefficients and Bootstrapped Standard Errors of LASSO Logistic Regression Model.

## 7 Conclusion

This analysis investigated how structural, accessibility, and spatial characteristics influence housing prices in Sindian District, New Taipei City. Using multiple linear regression and logistic regression models, we examined both continuous price prediction and classification of high-value properties.

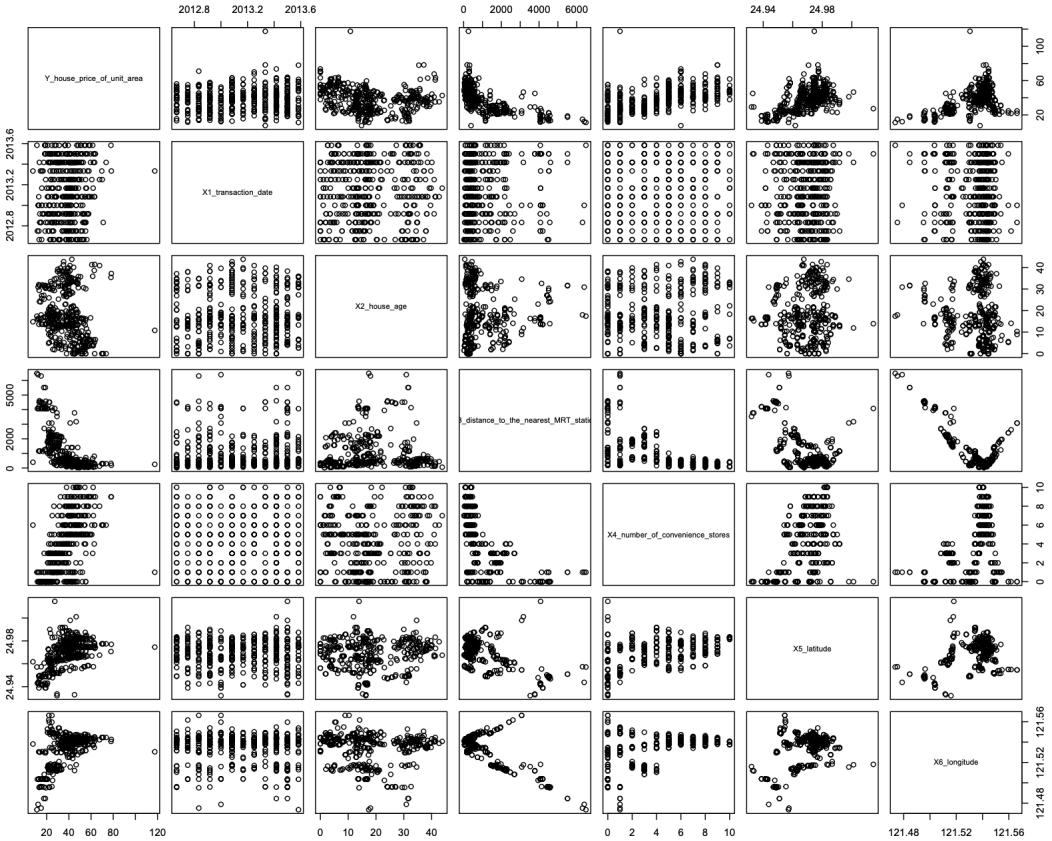
The results indicate that accessibility to public transportation and neighborhood amenities play an important role in determining housing prices. Properties located closer to MRT stations and in areas with more convenience stores tend to have higher values. Additionally, spatial location captured through principal component analysis highlights geographic variation in property values within the district.

Feature engineering techniques, including nonlinear transformations and spatial dimensionality reduction, significantly improved model performance and reduced violations of regression assumptions. Bootstrap analysis further demonstrated that the estimated coefficients are relatively stable across resampled datasets.

Overall, the findings emphasize the importance of accessibility and spatial characteristics in urban housing markets and demonstrate how statistical modeling can provide insights into the factors driving housing prices.

While the refined model improves fit and interpretability, several limitations remain. The dataset is relatively small and restricted to a single district, which may limit the generalizability of the results. Additionally, unobserved factors such as building quality, neighborhood demographics, and local policies may also influence housing prices but are not captured in the dataset.

# A First Appendix




Appendix A shows the pairwise scatter plots between all the variables in the dataset. House price per unit area shows a strong negative relationship with distance to the nearest MRT station. Prices drop quickly as the distance increases for houses close to MRT stations and then seem to level off far away from the MRT stations. There also seems to be a positive, linear relationship between house price and the number of nearby convenience stores. Latitude and longitude have a strong correlation, which makes sense, but this suggests potential collinearity if both variables are used to fit a regression model. House age has a noticeable negative relationship with house price. The transaction date doesn't show a strong relationship with house price or other predictors, but this could be due to the limited time frame. Distance to the nearest MRT station and house price per unit area, have extreme values that could be potential outliers.

## B Second Appendix

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                     y1   R-squared:                       0.582
Model:                            OLS   Adj. R-squared:                  0.576
Method:                 Least Squares   F-statistic:                     94.60
Date:                Sat, 07 Mar 2026   Prob (F-statistic):           4.83e-74
Time:                        21:50:57   Log-Likelihood:                -1487.0
No. Observations:                 414   AIC:                             2988.
Df Residuals:                     407   BIC:                             3016.
Df Model:                           6
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const      -1.444e+04   6775.386     -2.132      0.034   -2.78e+04   -1122.863
x1             5.1490      1.557      3.307      0.001       2.088       8.210
x2            -0.2697      0.039     -7.000      0.000      -0.345      -0.194
x3            -0.0045      0.001     -6.250      0.000      -0.006      -0.003
x4             1.1333      0.188      6.023      0.000       0.763       1.503
x5           225.4701     44.566      5.059      0.000     137.862     313.078
x6           -12.4291     48.581     -0.256      0.798    -107.930      83.072
==============================================================================
Omnibus:                      231.615   Durbin-Watson:                   2.153
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             3598.692
Skew:                           2.026   Prob(JB):                         0.00
Kurtosis:                      16.864   Cond. No.                     3.72e+07
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 3.72e+07. This might indicate that there are
strong multicollinearity or other numerical problems.
```

The original model explains a moderate portion of the variability in the response variable, with an $R^2$ of 0.582 and an adjusted $R^2$ of 0.576. The overall model is statistically significant, as indicated by the F-statistic of 94.60 with a p-value of 4.8310-74. Most predictors are statistically significant at conventional levels. Variables x1, x2, x3, x4, and x5 all have p-values below 0.01, indicating strong evidence of association with the response variable. The estimated coefficients for x1, x4, and x5 are positive, while the rest are negative. The variable x6 is not statistically significant (p = 0.798), indicating that there is insufficient evidence that it has a meaningful effect on the response variable in this model.

## C Third Appendix

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                     y1   R-squared:                       0.656
Model:                            OLS   Adj. R-squared:                  0.651
Method:                 Least Squares   F-statistic:                     129.6
Date:                Thu, 05 Feb 2026   Prob (F-statistic):           3.37e-91
Time:                        10:01:33   Log-Likelihood:                -1446.5
No. Observations:                 414   AIC:                             2907.
Df Residuals:                     407   BIC:                             2935.
Df Model:                           6
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const      -1.379e+04   2858.188     -4.825      0.000   -1.94e+04   -8171.725
x1             6.8888      1.420      4.851      0.000       4.097       9.680
x2            -0.8600      0.139     -6.169      0.000      -1.134      -0.586
x2_sq          0.0159      0.003      4.731      0.000       0.009       0.023
x3_log        -5.2678      0.591     -8.915      0.000      -6.429      -4.106
x4             0.4984      0.189      2.640      0.009       0.127       0.869
spatial_pc1    2.9720      0.448      6.635      0.000       2.092       3.853
==============================================================================
Omnibus:                      268.909   Durbin-Watson:                   2.125
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             6447.307
Skew:                           2.334   Prob(JB):                         0.00
```

The refined model explains a substantial portion of the variability in the response variable, with an $R^2$ of 0.656 and an adjusted $R^2$ of 0.651. The overall regression is statistically significant, as indicated by the F-statistic of 129.6 with a p-value of 3.3710-91. All predictors in the model are statistically significant at conventional levels. Variables x2, log(x3+1), and the intercept have negative estimated coefficients while the rest have positive ones.